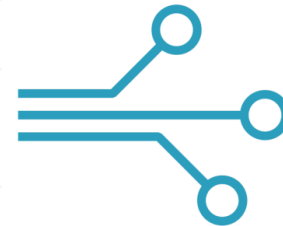




Java FX简介

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Java桌面应用GUI框架演进的历程



从Java 1.0开始，JDK中就提供了相应的开发框架，用于构建有可视化界面的Java桌面应用程序。

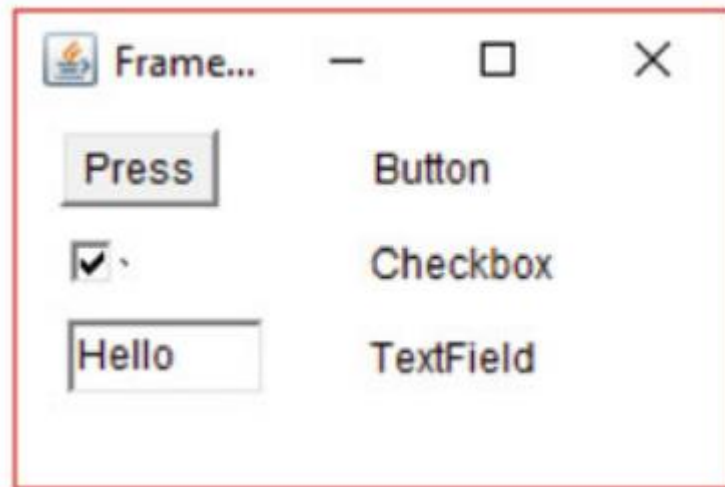


早期的GUI应用程序开发框架，叫AWT，后来，Sun公司推出了AWT的替代技术——Swing，AWT/Swing在很长的一段时间内都是Java平台主流的GUI开发技术。



Java更新一代的GUI应用程序开发框架，叫JavaFX，可以看成是AWT与Swing的替代者。

Java 桌面应用技术的演化



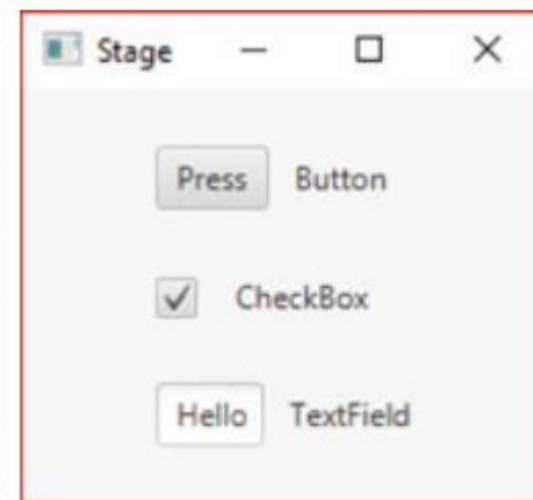
AWT

依赖底层操作系统绘制



Swing

Java自己负责绘制



JavaFX

利用GPU加速

早期Java GUI技术存在的问题

AWT:

write once, run anywhere



不同平台的UI控件，在用户交互特性上有许多微妙的差别

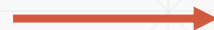
write many times, debug everywhere

Swing:

自己绘制所有的界面控件

用户的抱怨:

太慢

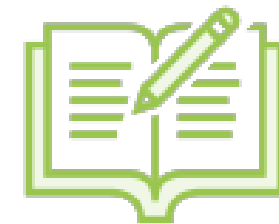


太丑



JavaFX

什么是Java FX?



JavaFX是Java所提供的的一个用于构建可视化用户界面（GUI）桌面应用程序的开发框架，是原有的AWT和Swing框架的取代者。

JavaFX可以全部使用代码生成UI界面，也可以使用“视图+代码”的MVC架构，以实现UI与功能代码的隔离。它在底层透明地调用GPU的硬件加速功能（如果可用的话），以提供高性能的图形渲染。



与AWT/Swing相比，JavaFX跟上了现代PC端GUI应用技术发展的步伐，学会它之后，能触类旁通地掌握其他类似的GUI框架或界面开发技术（比如WPF和Android）。



GUI

Graphical user interface

JavaFX vs. Swing



JavaFX可以完全替换掉Swing框架.....

Feature	JavaFX	Swing
Property Bindings	JavaFX properties support binding, meaning you can listen for changes in their values.	Swing properties do not directly support binding.
Declarative Layout	JavaFX has support for declarative layout via FXML.	Swing has no built-in support for declarative layout.
Styling	JavaFX supports CSS based and code based styling.	Swing only supports code based styling.
WebView	JavaFX has a WebView that can render modern web pages.	Swing has no WebView.
Graphics	JavaFX uses vector based graphics.	Swing uses pixel based graphics.
3D Graphics	JavaFX has built-in support for 3D graphics.	Swing requires the Java 3D API for 3D graphics.
Concurrency API	JavaFX has a built-in concurrency API.	Swing has no built-in concurrency API.
Age of toolkit	JavaFX is newer.	Swing is older.
Included in Java SDK	JavaFX is not included from Java 11 and forward.	Swing is still included, but will probably be removed one day.

使用JavaFX开发应用



JavaFX可以开发跑在多种平台和设备上的应用程序，包括：PC应用（Windows/Linux/Mac）、手机（iOS/Android）、嵌入式平台（Raspberry Pi），也可以集成AI等现代技术开发智能化应用，JavaFX本身是完全开放的

特定操作系统
下的本地应用



JavaFX



跨平台跨设备
的Web应用

兼具Web应用和
本地应用的特性

JDK 9引发的变化



JDK 9的重大变化，是模块化，相应地，JavaFX被抽取成了单独的模块，但一开始仍随JDK一起提供。



从JDK 11开始，JavaFX从JDK中移除了，因此，基于JDK 11以上版本开发JavaFX应用，往往需要单独地下载和安装JavaFX SDK。



如果使用IntelliJ IDEA这样的工具开发JavaFX应用，则它提供有现成的模板用于创建JavaFX项目，即使不单独下载JavaFX SDK，也可以开发JavaFX应用。

JavaFX相关模块

Module Name	Description
<code>javafx.base</code>	Defines the base APIs for the JavaFX UI toolkit, including APIs for bindings, properties, collections, and events.
<code>javafx.controls</code>	Defines the UI controls, charts, and skins that are available for the JavaFX UI toolkit.
<code>javafx.fxml</code>	Defines the FXML APIs for the JavaFX UI toolkit.
<code>javafx.graphics</code>	Defines the core scene graph APIs for the JavaFX UI toolkit such as layout containers, application lifecycle, shapes, transformations, canvas, input, painting, image handling, and effects. It also defines APIs for animation, CSS, concurrency, geometry, printing, and windowing.
<code>javafx.media</code>	Defines APIs for playback of media and audio content.
<code>javafx.swing</code>	Defines APIs for the JavaFX/Swing interoperability, which allows mixing Swing and JavaFX components in the same application.
<code>javafx.web</code>	Defines APIs for the loading and displaying web contents.

JavaFX社区官网

<https://openjfx.io>

此网站上包容新版
JavaFX的相关文档、
教程和资源。



JavaFX的源码

<https://github.com/openjdk/jfx>

The screenshot shows the GitHub repository page for `openjdk/jfx`. At the top, the repository name is displayed with icons for watching (54), starring (744), and forking (132). Below this is a navigation bar with tabs for Code, Pull requests (27), Actions, Security, and Insights. The main content area shows the `master` branch selected, with 3 branches and 565 tags. A commit by `kevinrushforth` is highlighted, with a description "Switch FX build to use JDK 15 as boot JDK" and 11,669 commits. Below the commit list, three folders are shown: `.idea`, `.jcheck`, and `apps`, each with a corresponding commit message and date. On the right side, there is an "About" section with a link to the project page and tags for `openjdk`, `javafx`, and `java`. At the bottom right, there are links for "Readme" and "View license".

openjdk / jfx

Watch 54 Star 744 Fork 132

Code Pull requests 27 Actions Security Insights

master 3 branches 565 tags

Go to file Add file Code

kevinrushforth 8253123: Switch FX build to use JDK 15 as boot JDK 47e67b4 12 hours ago 11,669 commits

.idea	8234174: Change IDEA VCS mapping to Git	10 months ago
.jcheck	8240499: Enforce whitespace checking for additional source files	3 days ago
apps	8238434: Ensemble: Update version of Lucene to 7.7.2	7 months ago

About

<https://openjdk.java.net/projects/openjfx/>

openjdk javafx java

Readme

View license